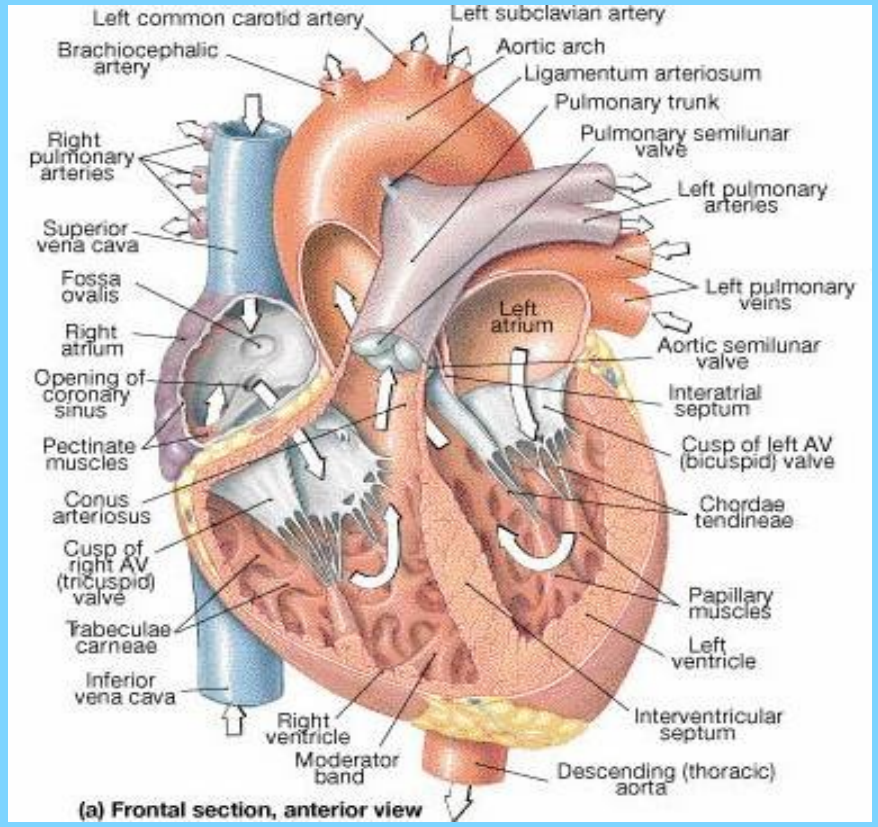
Chambers and internal features of heart

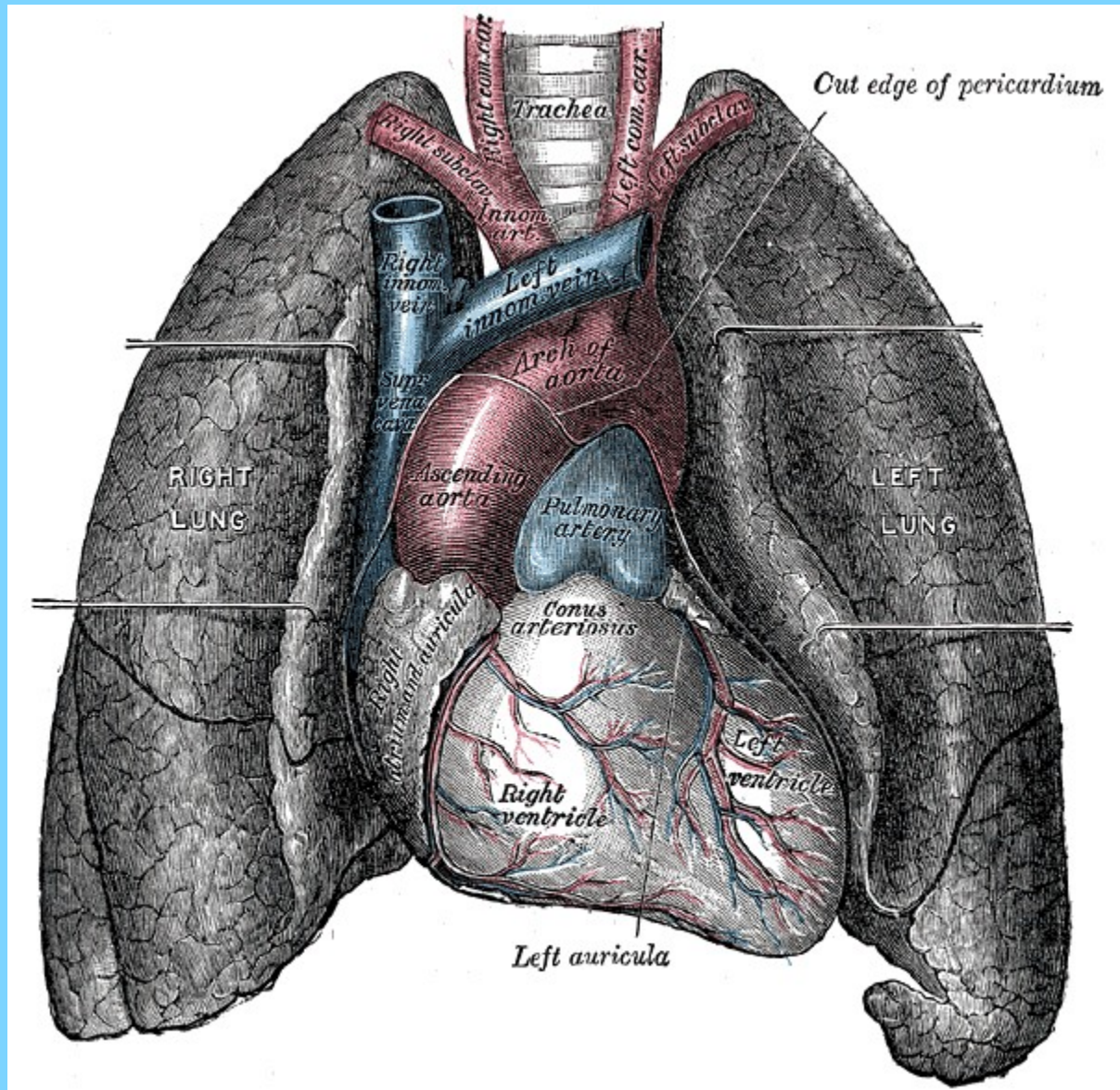


Learning objectives

- At the end of the lecture the student should know the following
- Chambers of the heart
- Internal structures of the heart
- Valves of the heart

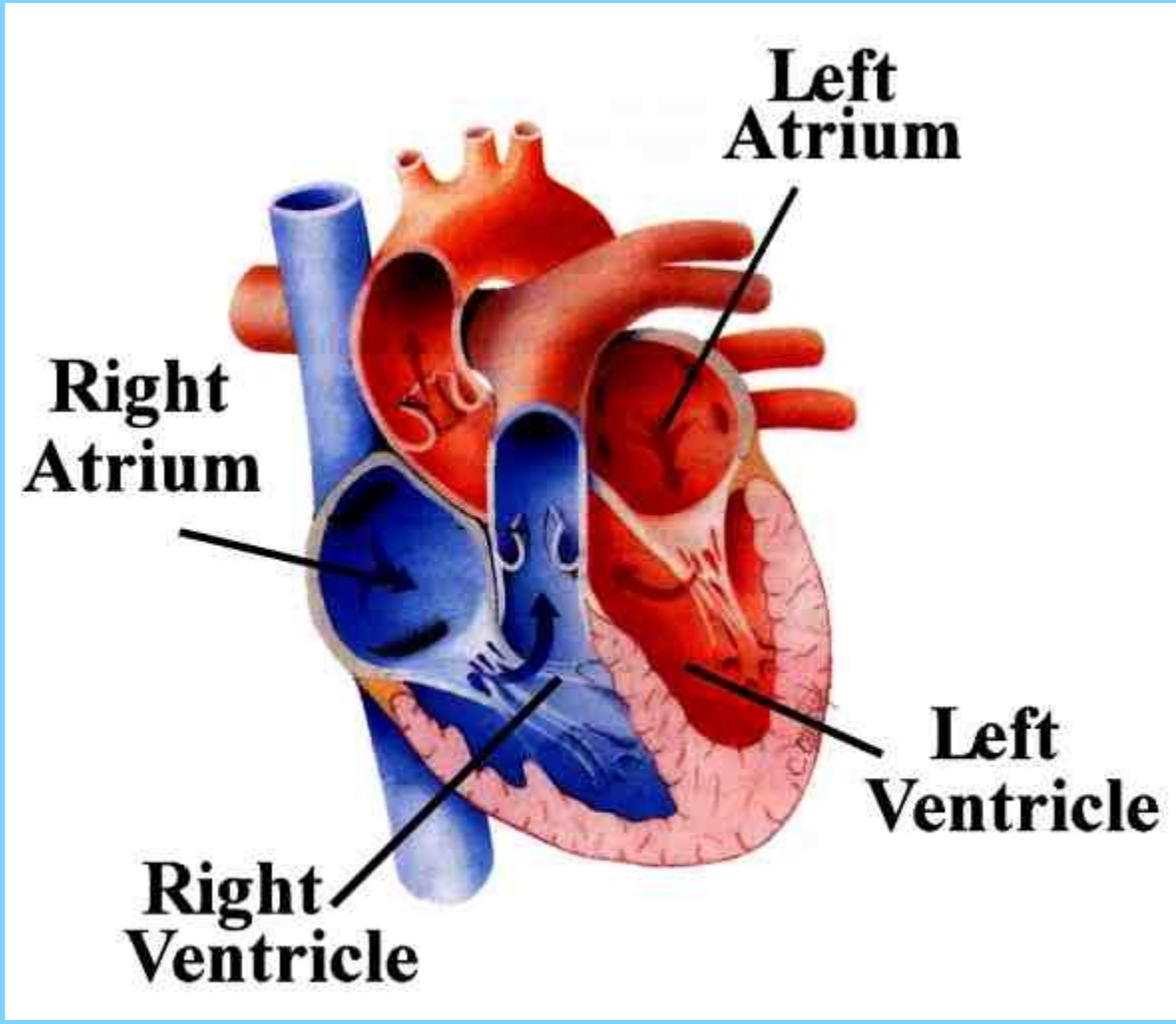
Heart

- Heart is a muscular organ.
- Lies between the lungs in the middle mediastinum
- One-third is on the right
- Two third in the left



Chambers of heart

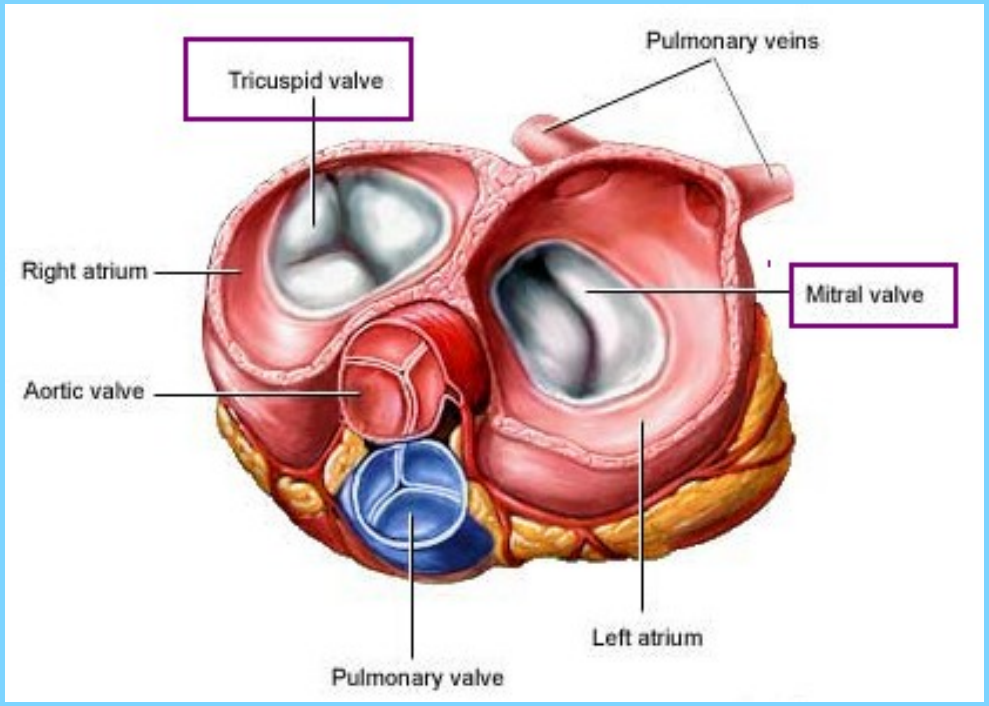
- Divided by septa into right and left halves
- Subdivided into atrium and ventricle
- So the heart has four chambers
- Right atrium
- Left atrium
- Right ventricle
- Left ventricle



Valves

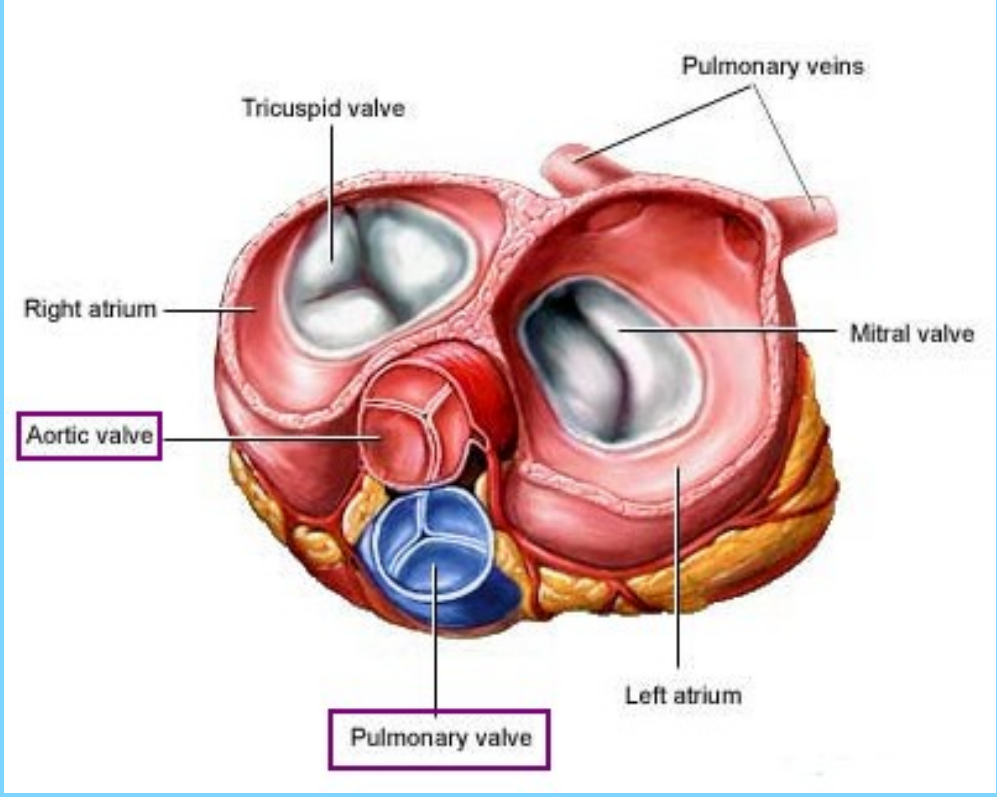
- **Valves are flaps of tissue that stop blood flowing backwards**
- They are two kinds

  The first kind is the massive **atrio-ventricular valves**, (AV valves) that prevent blood in the ventricles from flowing back into the atria.
- The flaps of these valves are attached to the walls of the ventricles by tendons – chordae tendinae



Valves

- The second kind of valve is pocket shaped flaps of tissue called the **semilunar** (half moon) **valves**
- They are called the **pulmonary and aortic valves** and found at the entrance of the pulmonary artery and aorta respectively

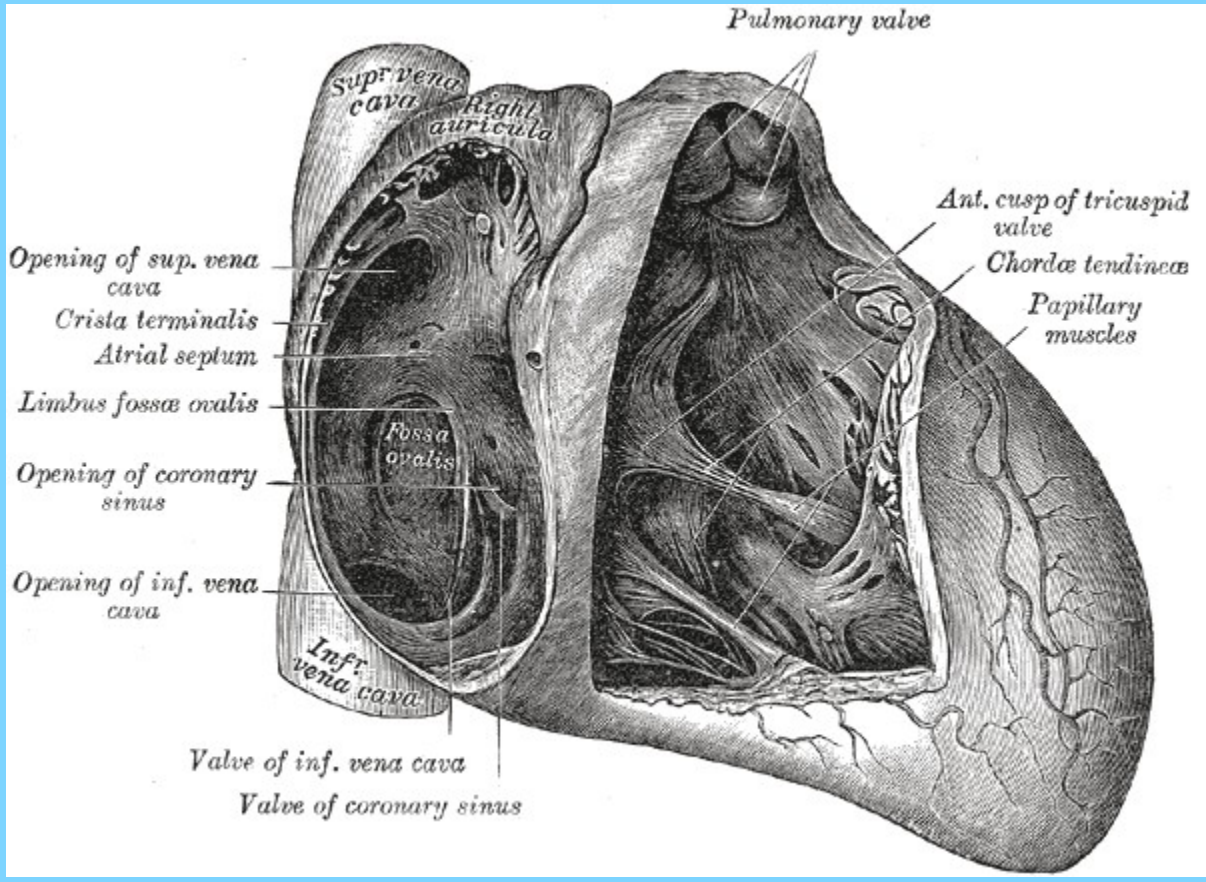


Right atrium

- Larger than left
- 57c.c capacity
- Consists of 2 parts

  Main cavity called sinus venarum

  Anterior small portion called auricula



Openings of right atrium

- Superior vena cava

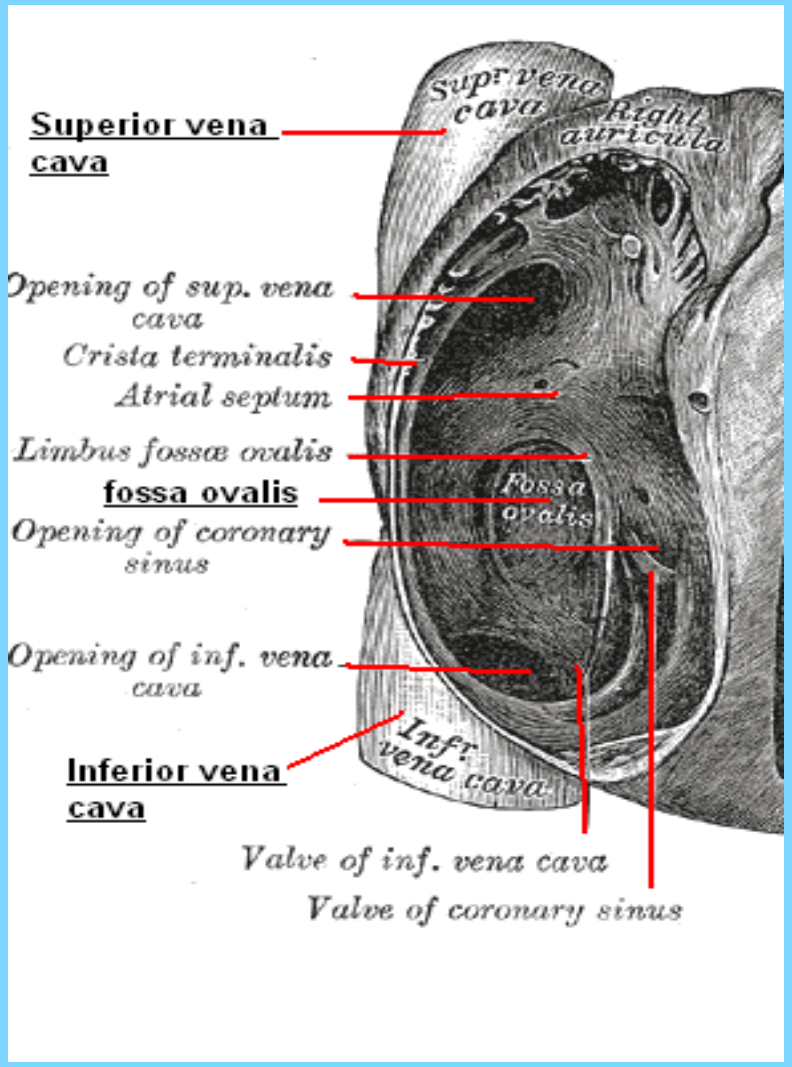
  Has no valve
- Inferior vena cava

  Eustachian valve
- Coronary sinus

  Btw the opening of inf. vena cava and AV opening

  Has a valve
- Foramina venarum

  Orifices of minute veins
- AtrioVentricular opening



Valves of right atrium

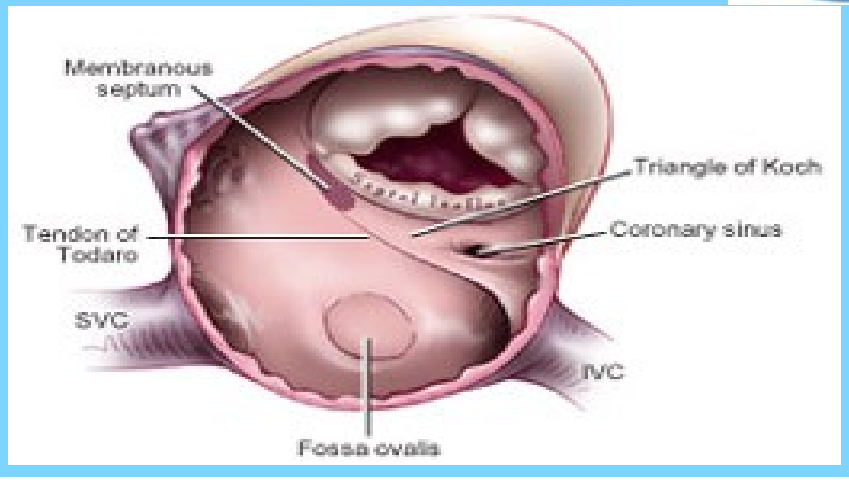
- **Valve of inferior vena cava**

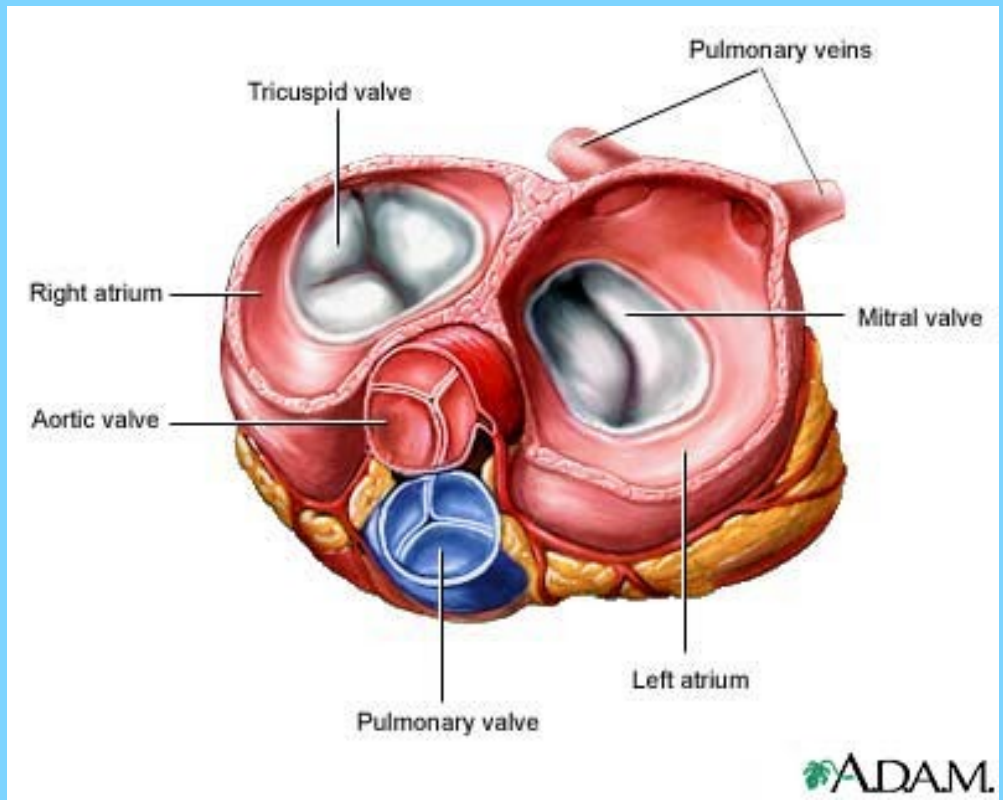
  Semilunar valve
- **Valve of coronary sinus**

  Semicircular fold
- **Fossa ovalis**

  Oval depression on the septal wall of the atrium
- **Tricuspid valve**

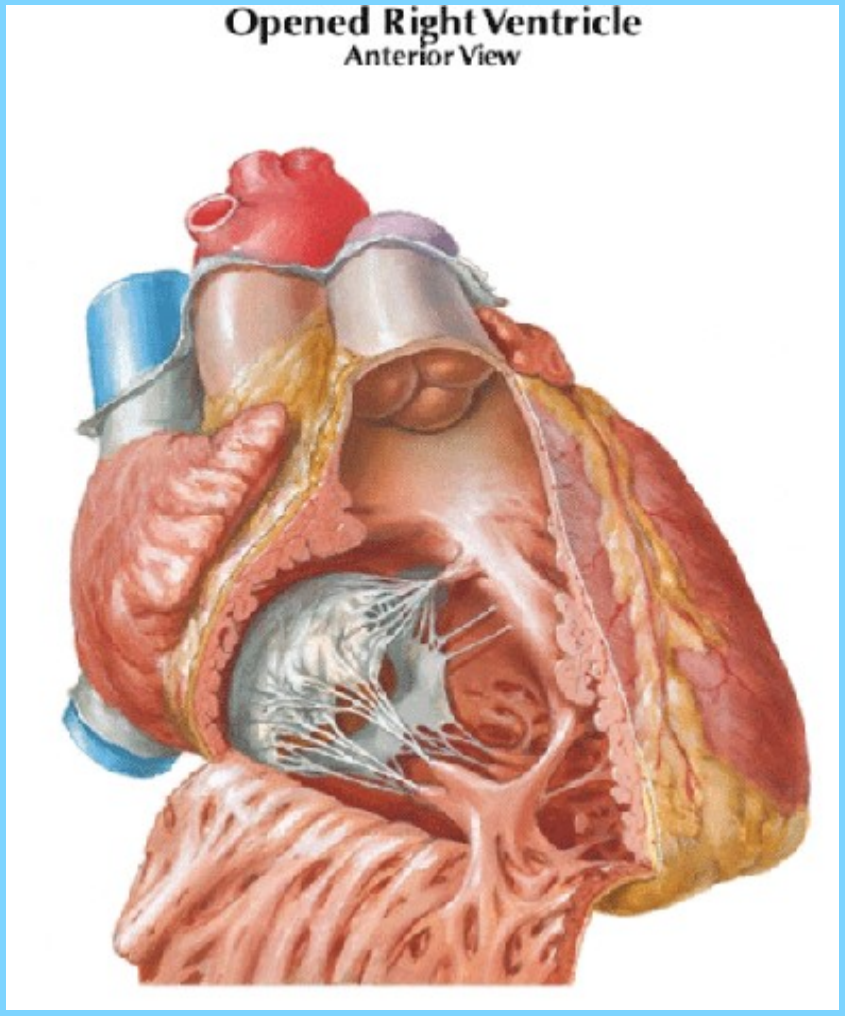
  Between the right atrium and right ventricle





Right ventricle

- 85c.c capacity
- Extends from right atrium to near the apex of the heart
- Inferior border rests on diaphragm
- Anterosuperior surface is post sternal
- Interventricular septum makes the Posterior surface.



Openings of right ventricle

- **Right atrioventricular orifice**

  Communication between right atrium and ventricle

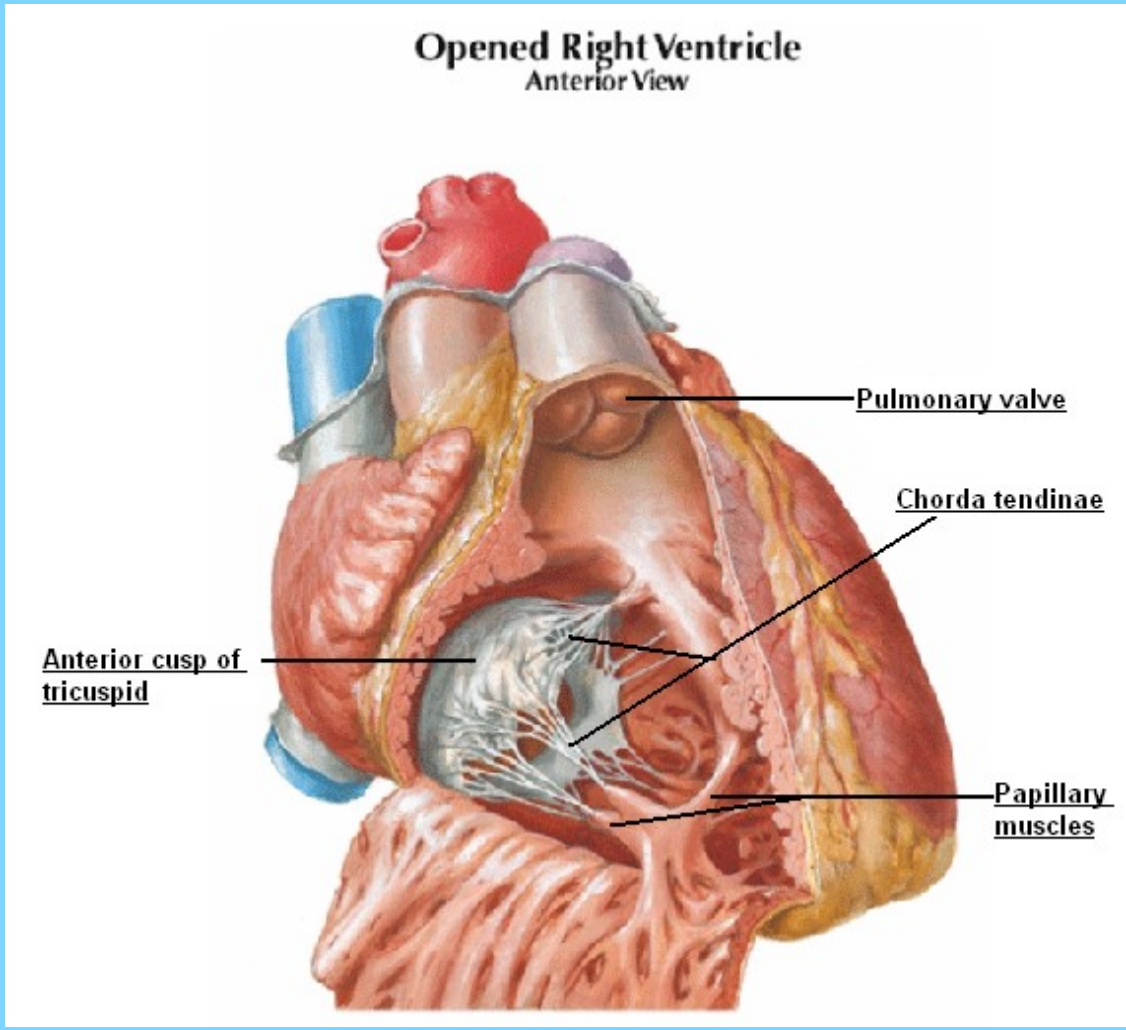
  4cm diameter

  Has a tricuspid valve

- **Pulmonary orifice**

  Circular in form

  Guarded by the pulmonary semilunar valves



Valves of right ventricle

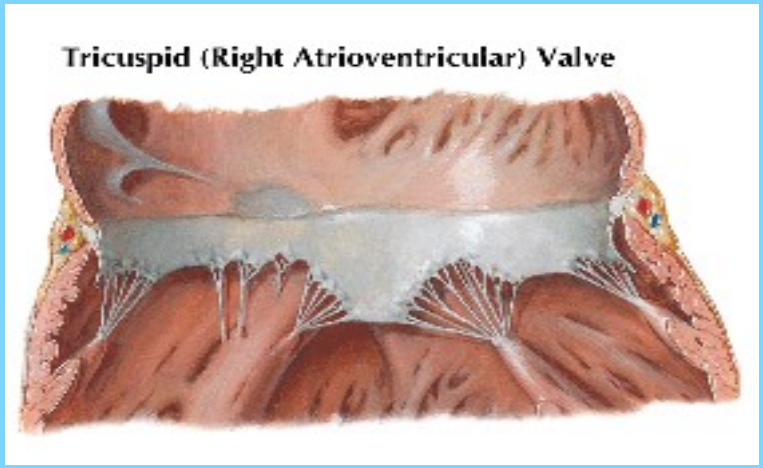
- **Tricuspid valve**

  Has 3 triangular cusps:

  Anterior or Infundibular cusp

  Posterior or Marginal cusp

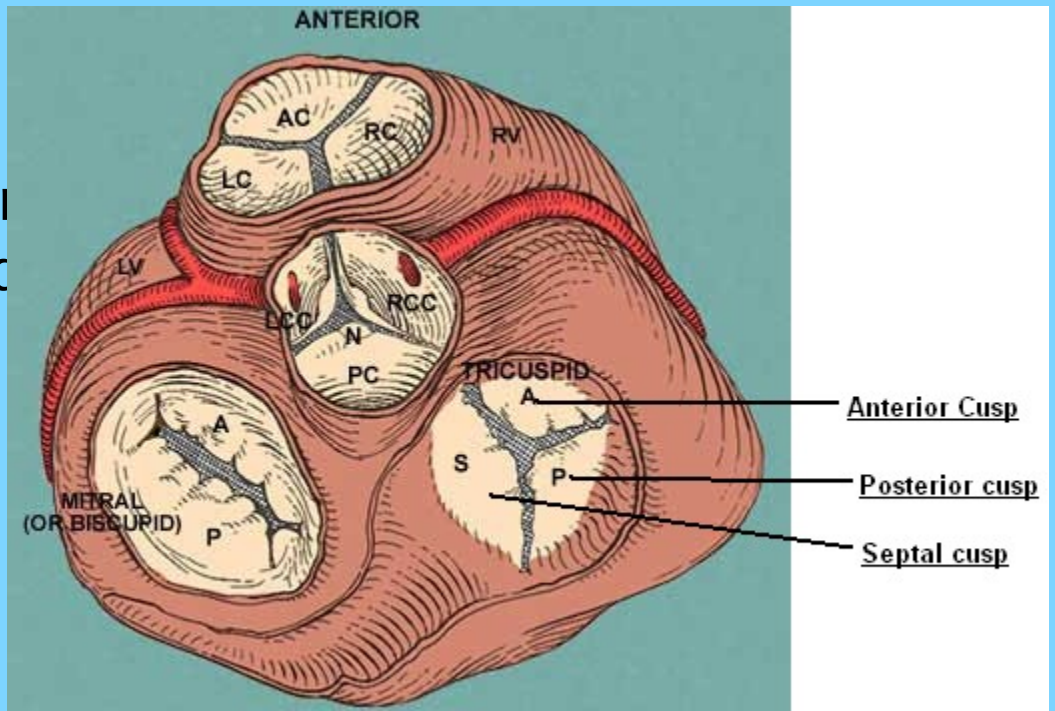
  Medial or Septal cusp



- **Pulmonary valve**

  3 in number

  Attached to the wall of arter
  at the junction of artery and
  ventricle

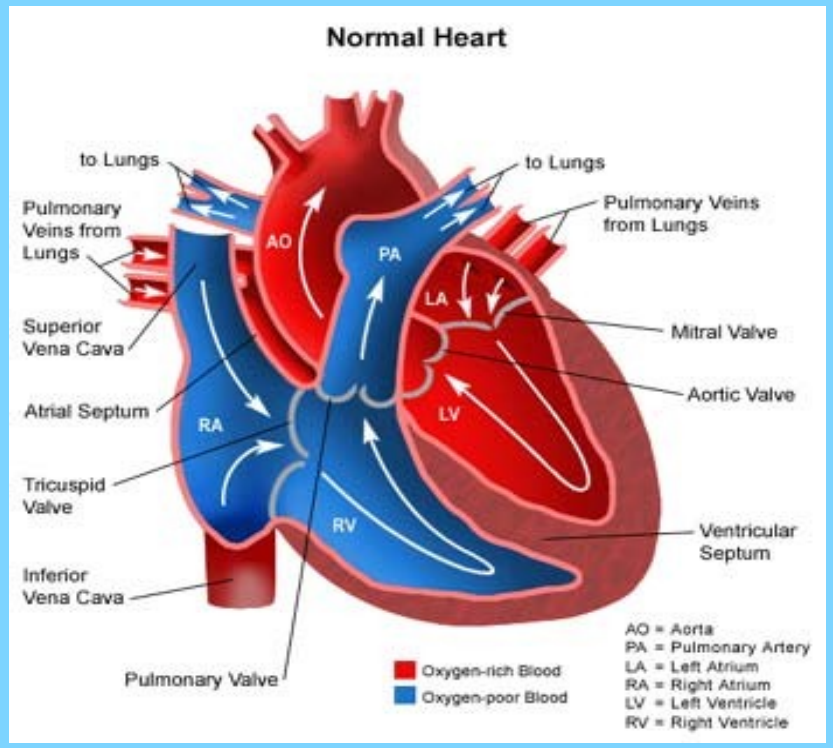


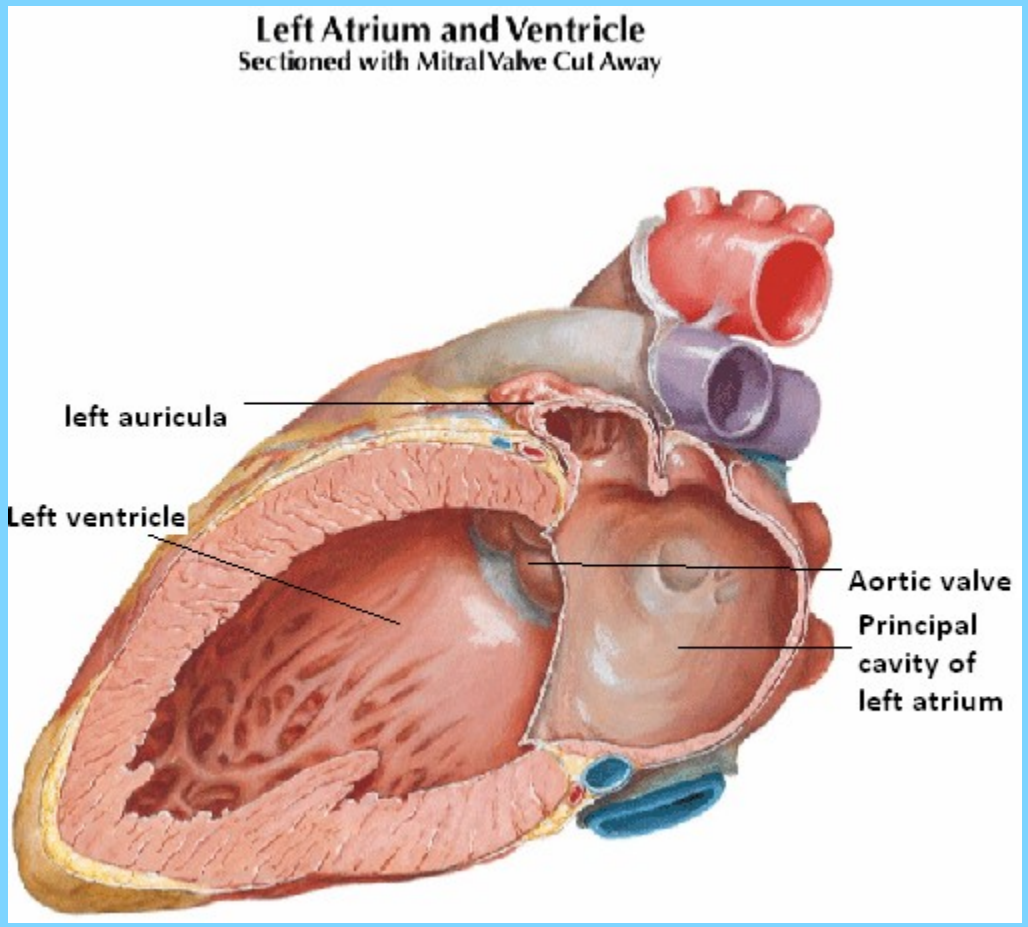
Left atrium

- Smaller than the right
- Walls are thick
- Consists of

  Principal cavity

  Auricula





Openings of left atrium

- **Pulmonary veins:**

  4 in number

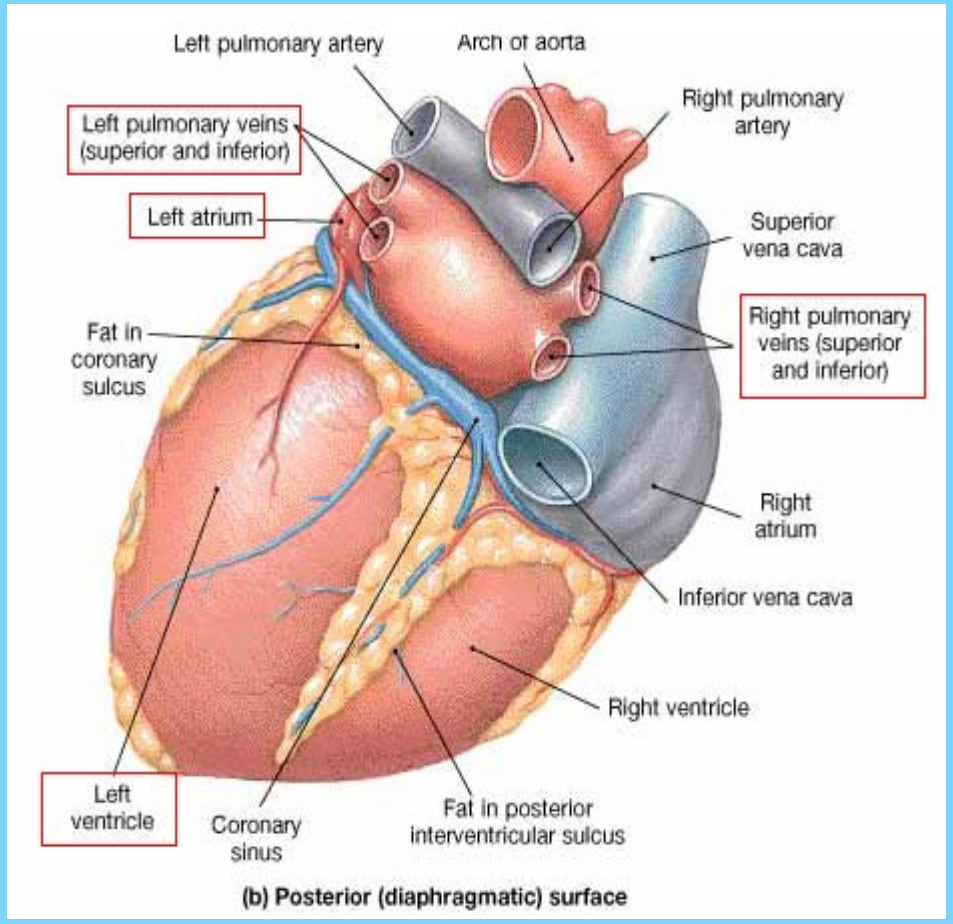
  Opens in the posterior surface

  Has no valves

- **Left Atrioventricular orifice**

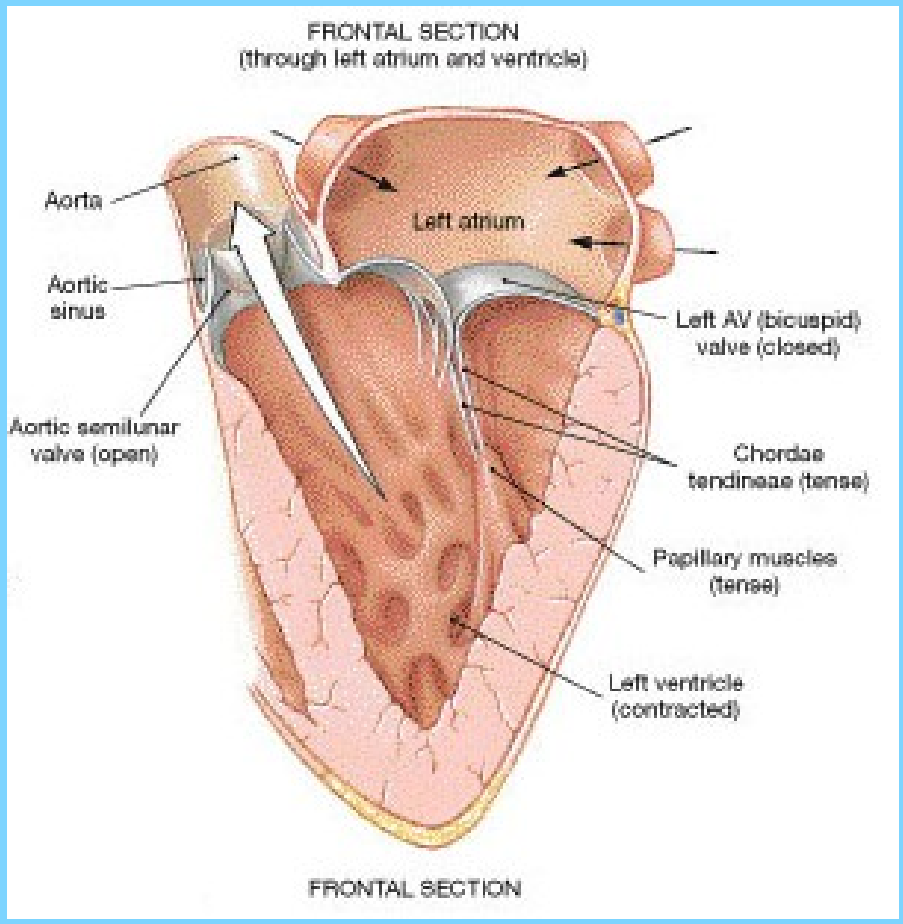
  Between left atrium and ventricle

  Has a mitral valve



Left ventricle

- Longer and three times thicker than right
- Conical in shape
- Forms the apex of the heart



Openings of left ventricle

- **Left Atrioventricular orifice**

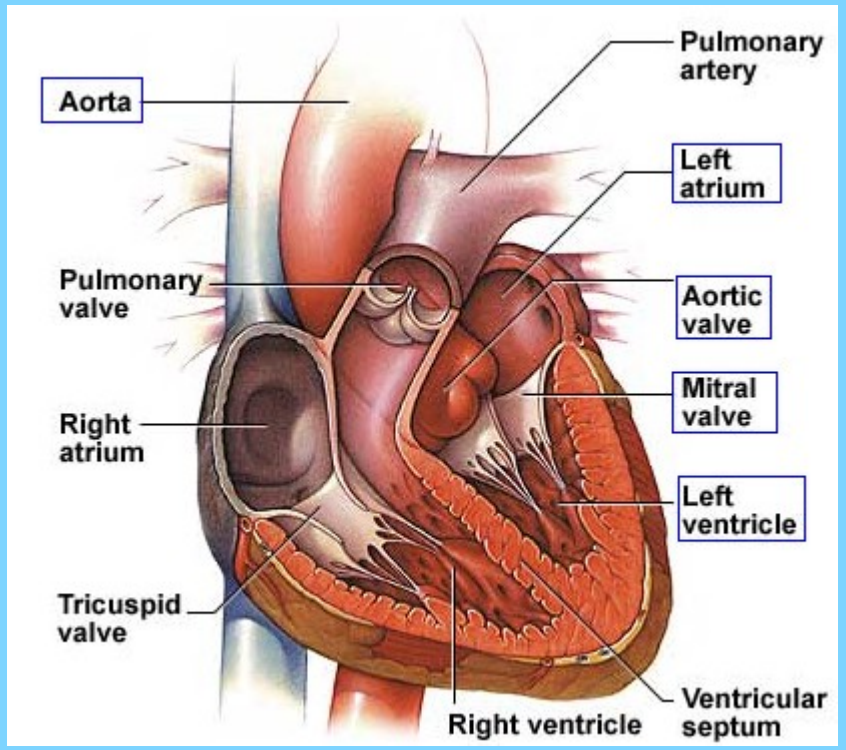
  Below and left to the aortic orifice.

  Has a mitral valve

- **Aortic orifice**

  Circular aperture

  Has an aortic semilunar valve.



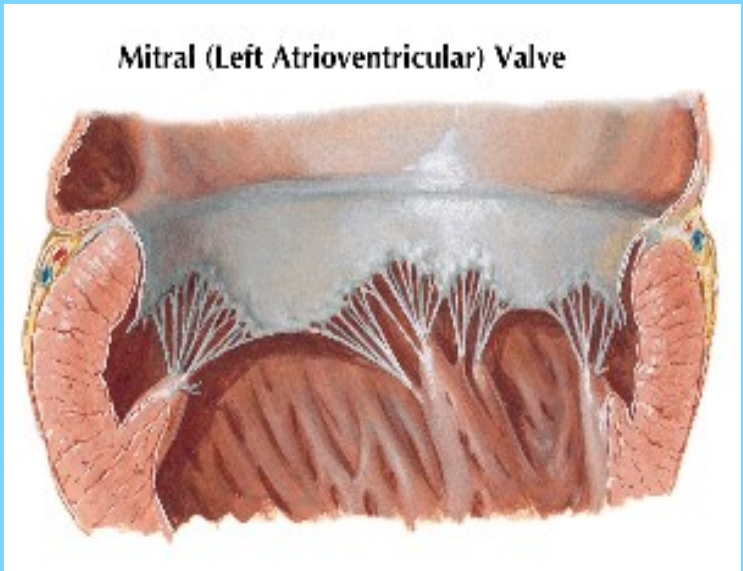
Valves of left ventricle

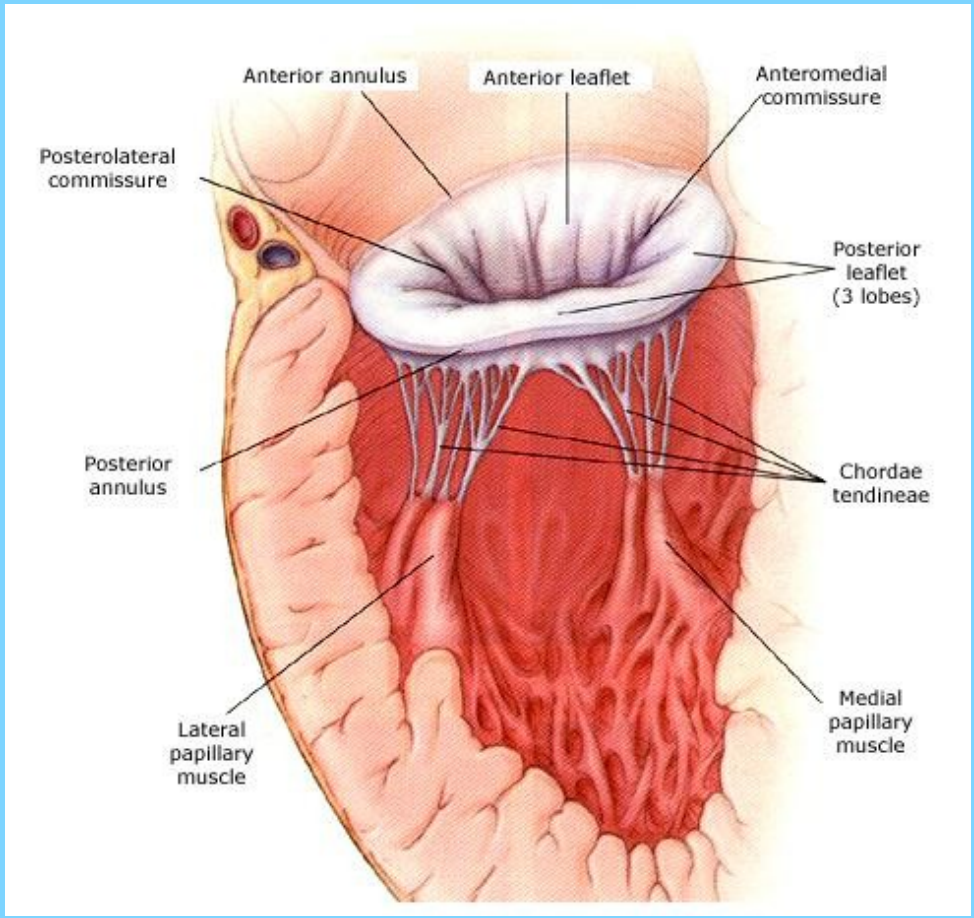
- **Bicuspid or mitral valve**

Consists of two cusps

Anterior or aortic cusp

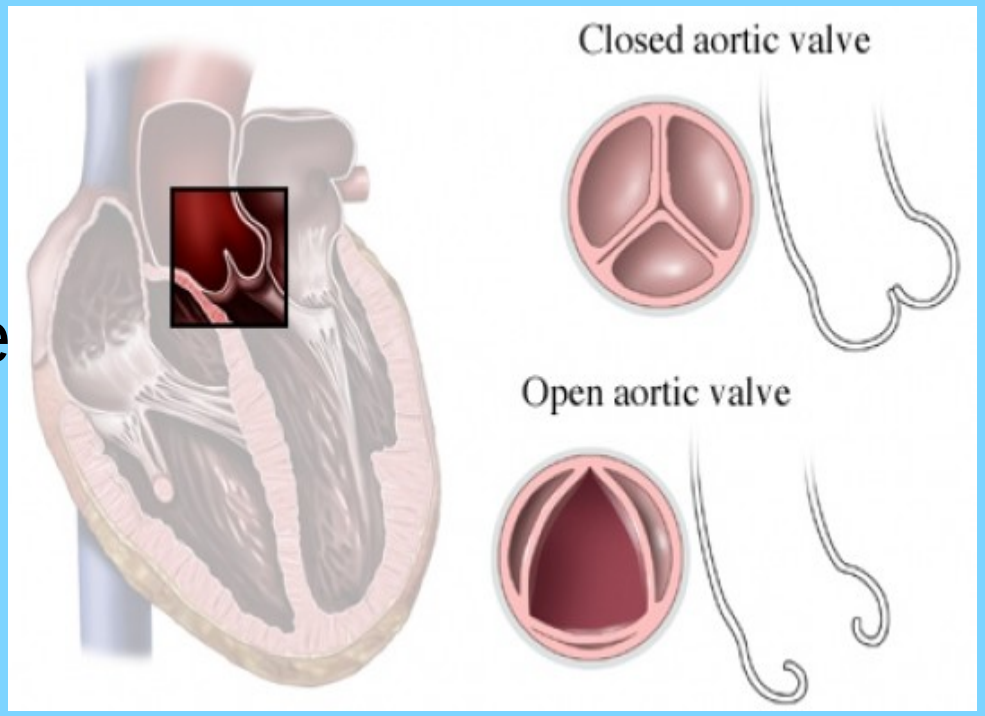
Posterior cusp





Valves of left ventricle

- **<u>Aortic semilunar valves</u>**
- Controls flow of blood out of the left ventricle to the aorta
- Three in number
- Two anterior and one posterior



Internal structures of ventricle

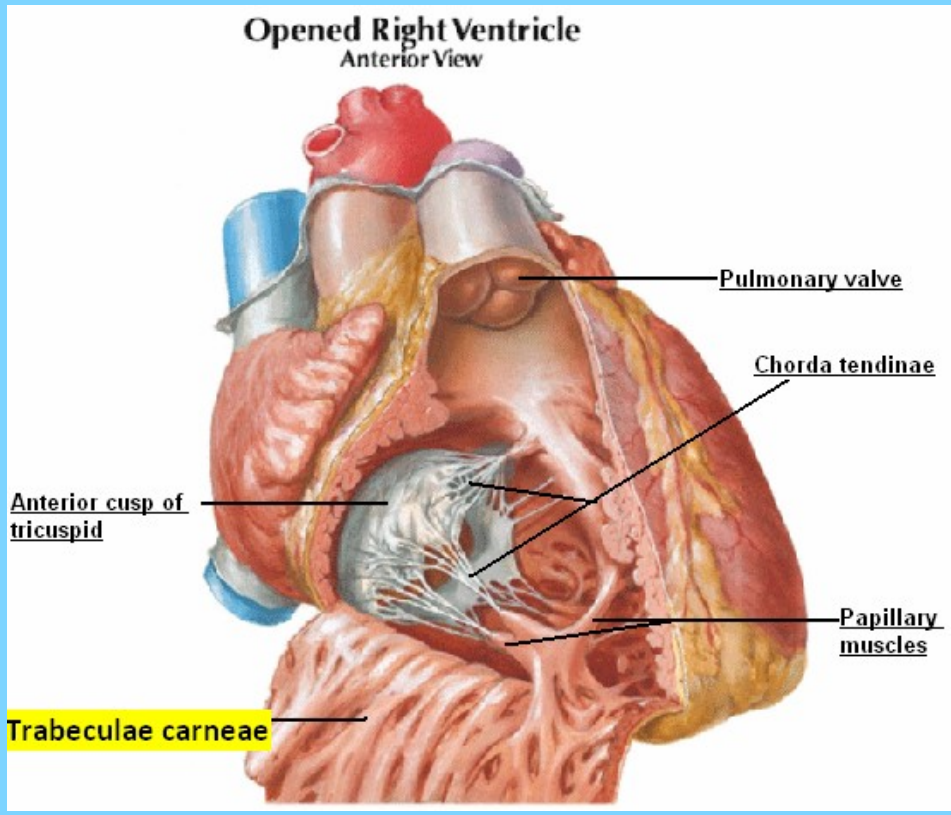


- **<u>Trabeculae carneae</u> (fleshy beams)**

  They are rounded or irregular muscular columns which project from the whole of the inner surface of the ventricle, with the exception of the conus arteriosus.

- **<u>Three types</u>**

1. Fixed along their entire length
2. Fixed at edges free in the middle.
3. Papillary muscles



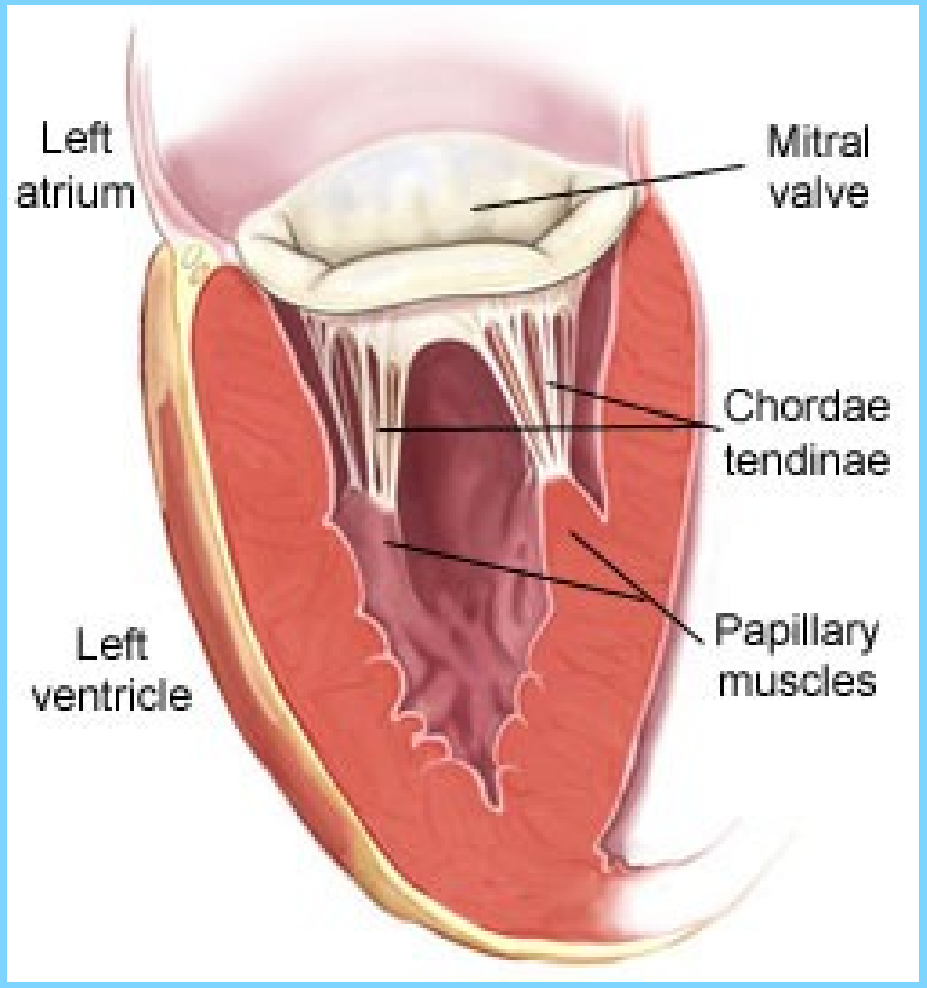
Internal structures of ventricles

- **Papillary muscles**

  Muscles that are attached to the AV valve cusps to limit the movement of the mitral and tricuspid valves

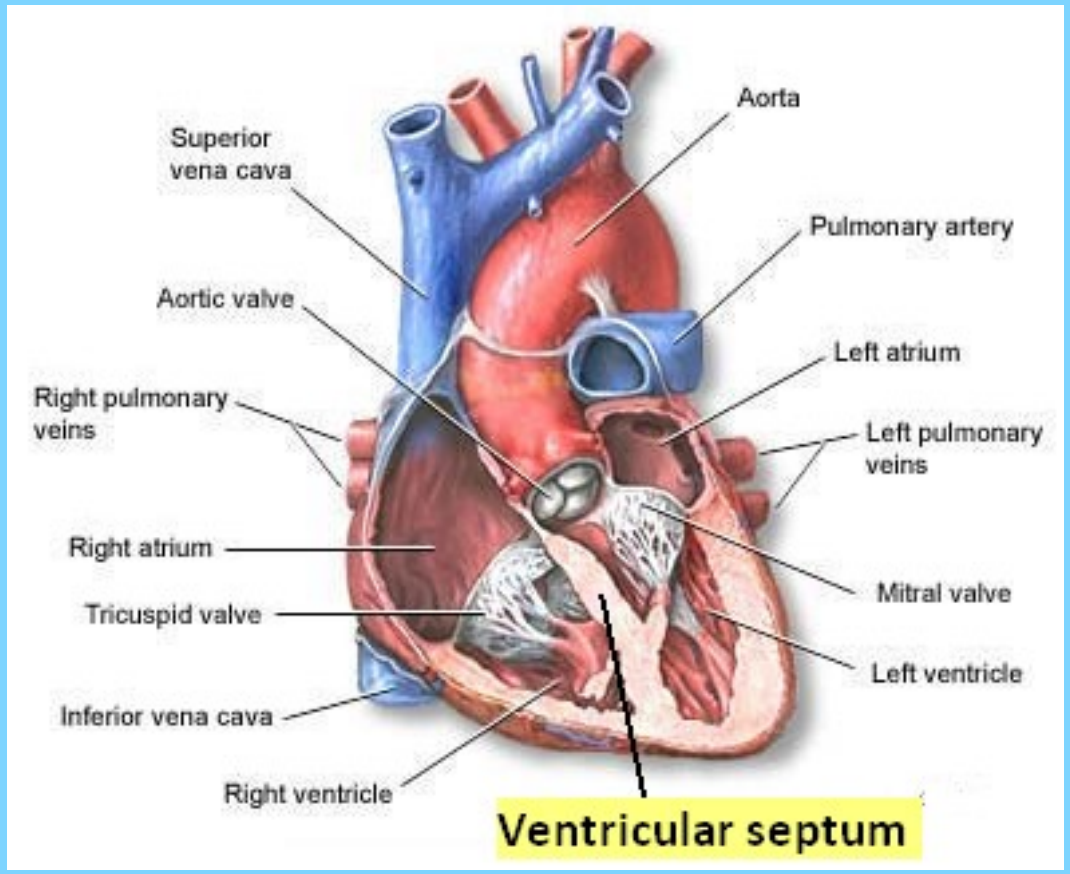
- **Cordae tendineae**

  cord-like tendons that connect the papillary muscles to the tricuspid valve and the mitral valve in the heart.



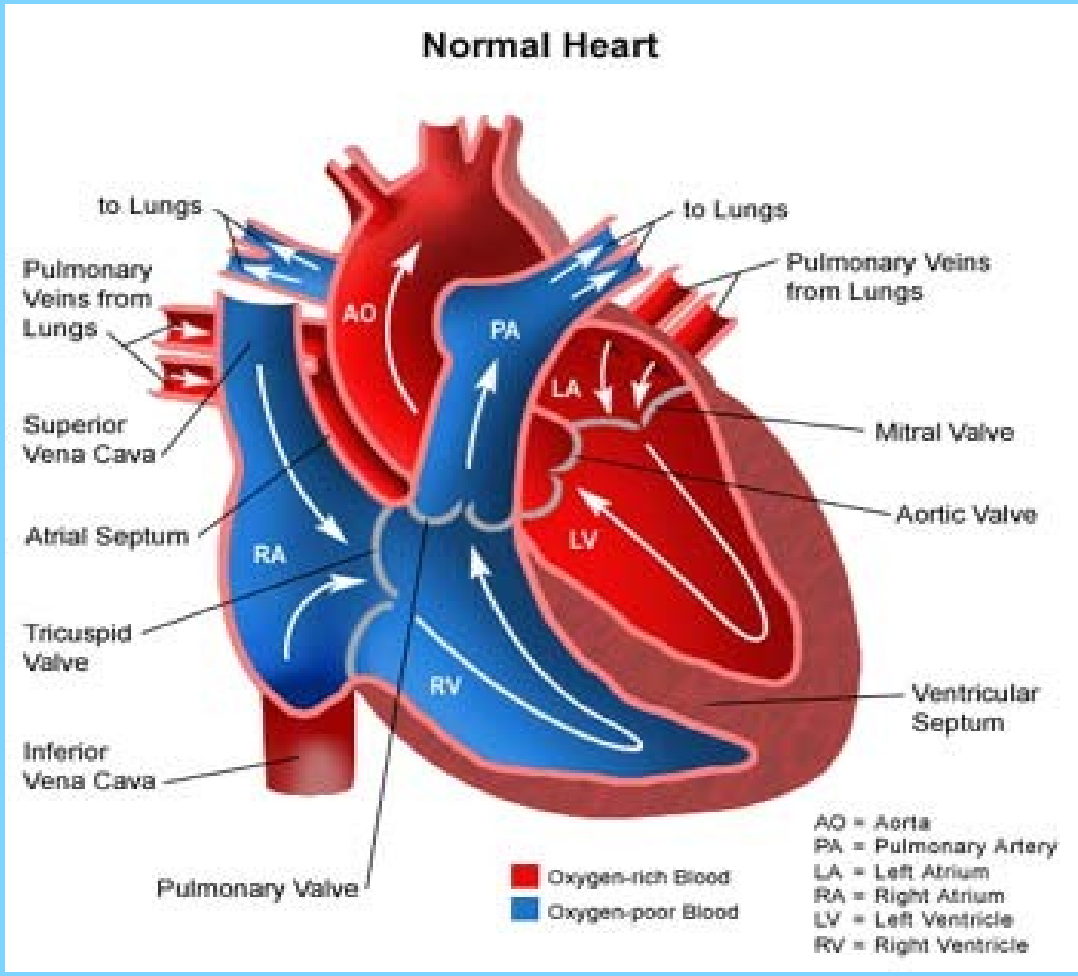
Ventricular septum

- The ventricular septum is wall separating the lower chamber of the heart from one another
- Directed obliquely backward and to the right, and is curved with the convexity toward the right ventricle:



Chambers of the Heart

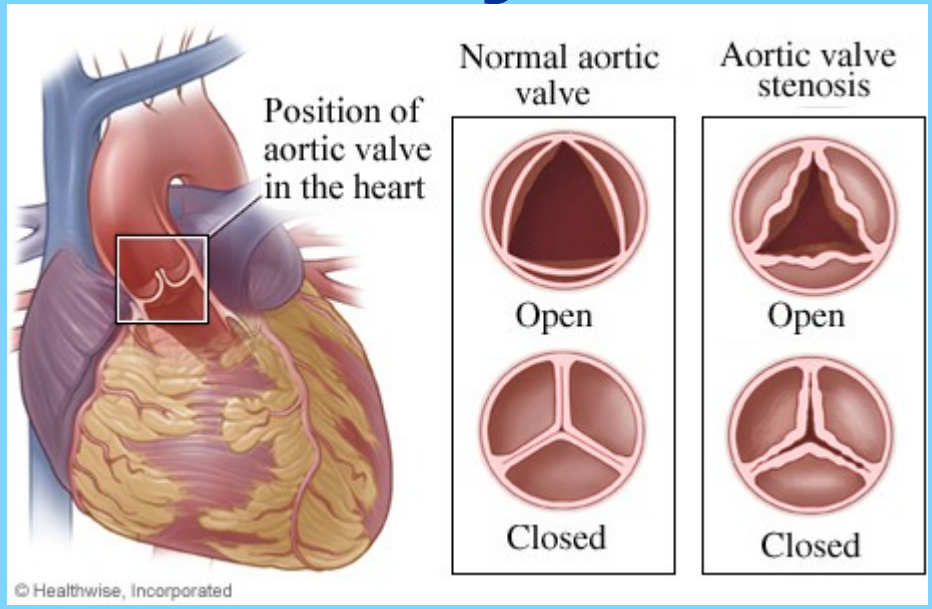
- The right atrium receives blood from the body.
- The right ventricle pumps blood to the lungs.
- The left atrium receives blood from the lungs.
- The left ventricle pumps blood to the rest of the body.



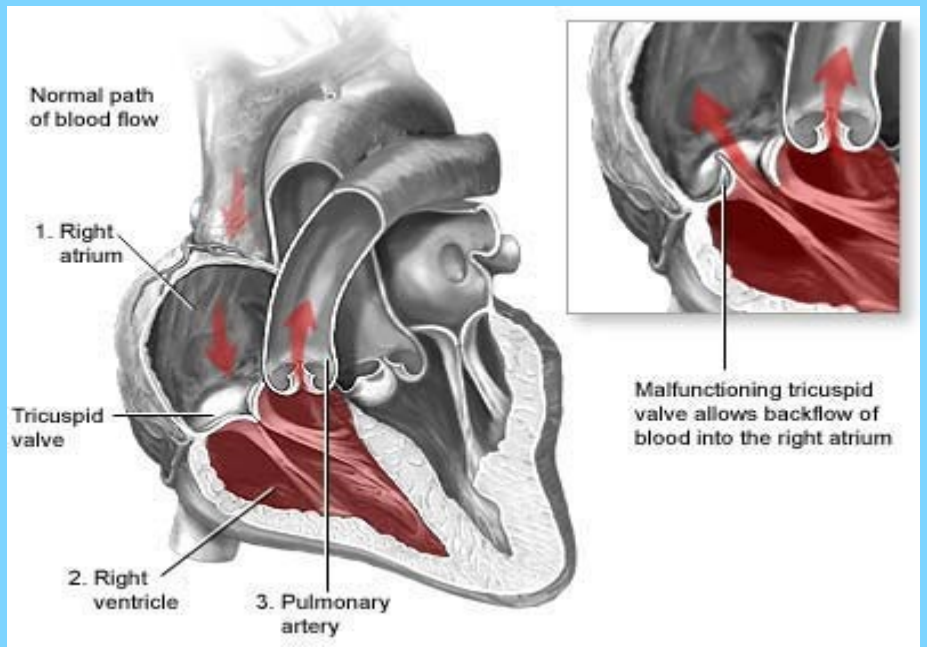
Applied anatomy

Aortic stenosis

Thickening and hardening of aortic valve which prevents the valve to open properly



Tricuspid stenosis



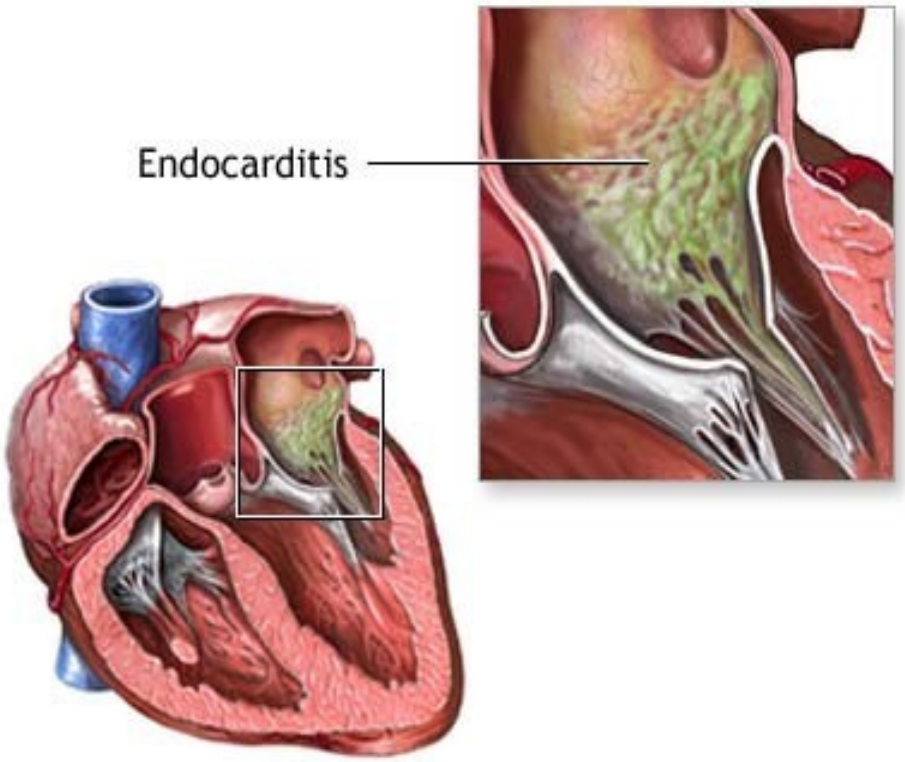
Applied anatomy

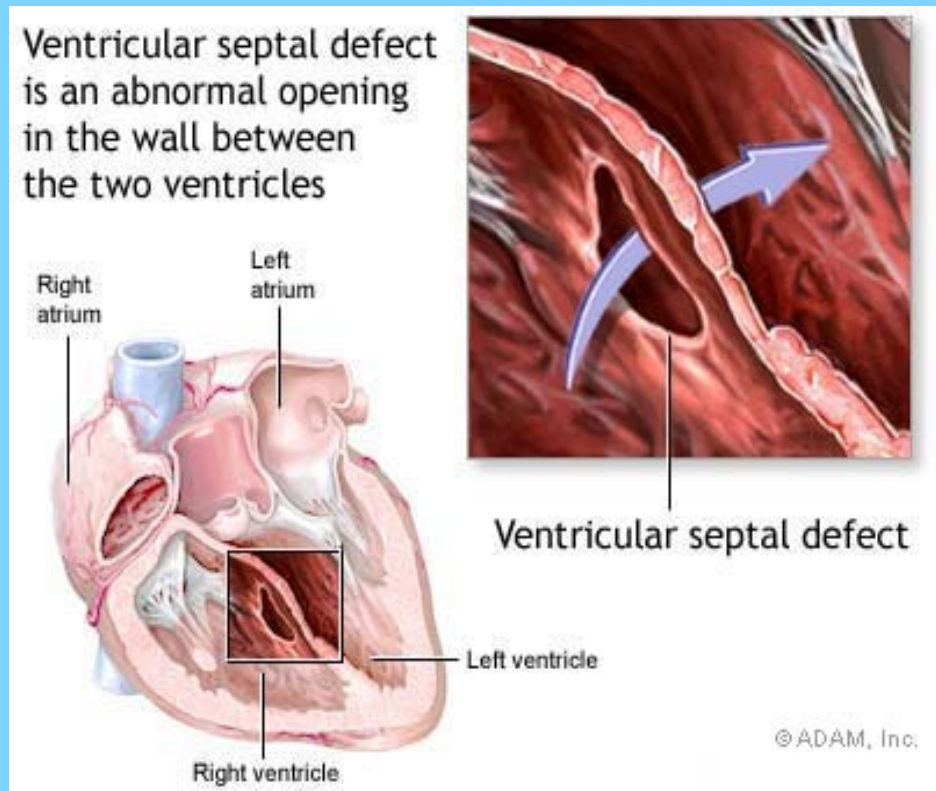
Ventricular septal defect

.is a hole (defect) in the wall that separates the lower chambers of the heart

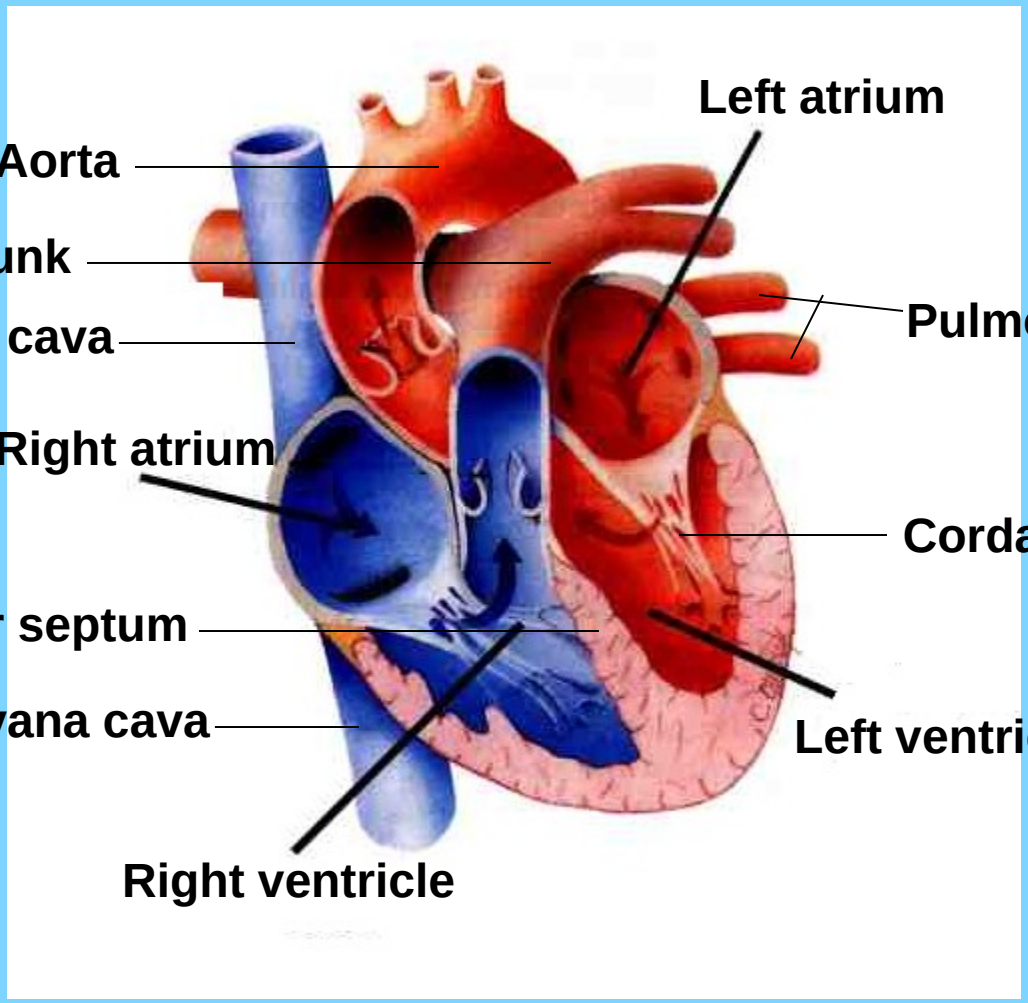
Endocarditis

Infection of the endocardium





??Identify


Left atrium
Aorta
Pulmonary trunk
Pulmonary veins
Superior vena cava
Right atrium
Cordae tendinae
Ventricular septum
Inferior vana cava
Left ventricle
Right ventricle

??Identify the structures

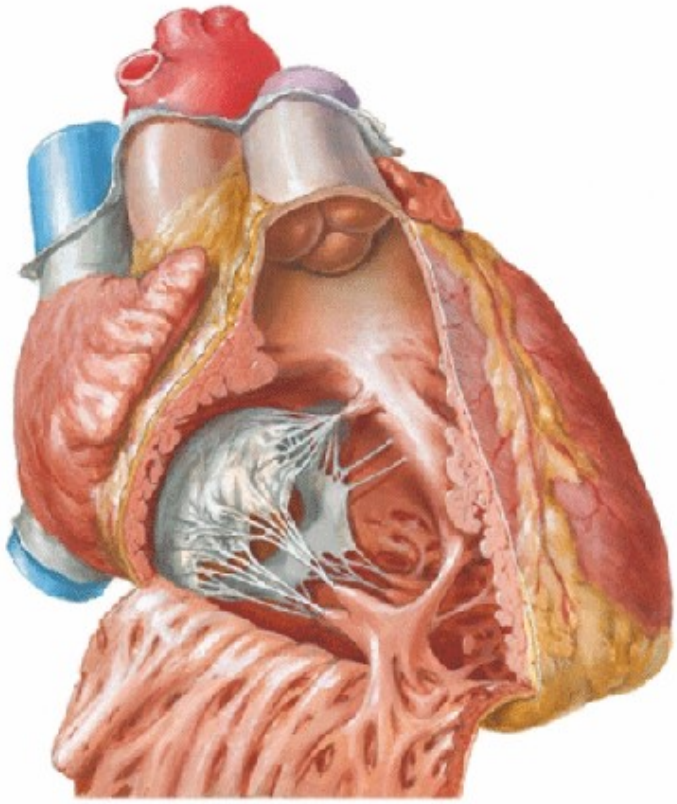
A



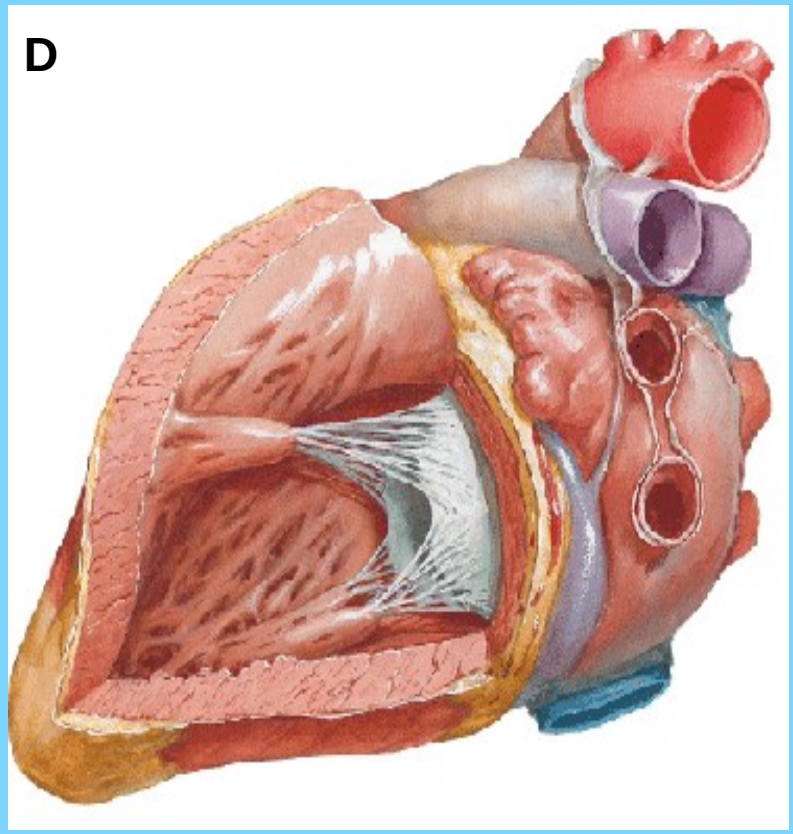
B



C



D





Thank you